Trader Behavior Insights Report

Analyzing the Relationship Between Market Sentiment and Trading Performance

1. Introduction

This report analyzes how trader behavior and performance on the Hyperliquid platform change based on Bitcoin market sentiment. By combining historical trader data with the Bitcoin Fear & Greed Index, the objective is to understand whether traders perform differently during periods of Fear versus Greed, and to identify behavioral patterns that may influence risk, profitability, and trading volume.

Two datasets were merged:

1. Hyperliquid historical trade data (tick-level trading records)
2. Bitcoin Fear & Greed Index (daily sentiment)

The unified dataset enables exploration of the relationship between sentiment and trader actions at the hourly, daily, and aggregate levels.

2. Data Overview and Preparation

The historical data includes detailed trade attributes such as execution price, position size, PnL outcomes, timestamps, and trade direction.
The sentiment dataset provides a daily classification of market mood, along with a numeric sentiment score between 0 and 100.

Key preprocessing steps included:

- Converting UNIX and IST timestamps into a unified datetime format
- Creating a clean date field for alignment with daily sentiment
- Standardizing categorical values (e.g., side, sentiment labels)

- Converting numeric columns such as size, trade value, and PnL
- Merging the datasets based on date
- Handling missing sentiment values using per-date forward/backward fill
- Creating engineered features such as win/loss indicators, hourly buckets, and relative trade size metrics

The final dataset enabled reliable comparison of trading behavior under different sentiment regimes.

3. Methodology

The following metrics were analyzed:

1. Profitability
    - Distribution of Closed PnL
    - Average PnL per sentiment category
2. Risk and Trading Behavior
    - Trade frequency
    - Trading volume (USD)
    - Hour-of-day activity patterns
3. Performance Metrics
    - Win rate across sentiment conditions
    - Stability and consistency of PnL during Fear vs Greed
4. Visual Analysis
   All analysis was supported by visualizations including bar charts, boxplots, heatmaps, and time-series plots.

Each chart was exported and saved to the outputs/ directory as required.

4. Key Findings

4.1 Trading Activity Patterns

A comparison of trade counts by hour and sentiment shows that trading activity is overwhelmingly higher during Fear. The busiest periods occur between 15:00 and 21:00, with Fear-driven trades dominating the dataset.
Greed sentiment exhibits minimal activity, suggesting that traders tend to react more strongly during periods of uncertainty and volatility.

4.2 Trading Volume

Total trading volume during Fear is significantly higher, approximately $420 million, compared to almost negligible volume during Greed. This indicates that traders deploy larger positions when market sentiment is negative, likely due to volatility presenting more opportunities for entry and exit.

4.3 Win Rate

Win rates show a substantial difference between sentiment regimes:

- Fear: approximately 40 percent
- Greed: approximately 14 percent

This suggests that traders perform more effectively during Fear, potentially due to clearer price movement, increased volatility, or more mean-reverting setups.

4.4 Profitability (PnL Distribution and Averages)

The distribution of Closed PnL highlights several important insights:

- Fear periods contain a dense cluster of frequent profitable trades along with noticeable positive outliers.
- Greed periods show fewer trades and fewer profitable outliers.

Average Closed PnL is slightly higher during Fear (around $90) than Greed (around $88). While the averages appear similar, they should be interpreted carefully:
Fear's average comes from a large and stable sample, while Greed's average is based on very few trades and is therefore less reliable.

4.5 Overall Behavioral Interpretation

The combined results indicate that Fear is a more active and more profitable environment for traders. High volatility during Fear leads to:

- More opportunities
- Higher participation
- Better win rates
- Greater trading volume

In contrast, Greed regimes show limited action, less favorable outcomes, and weaker performance metrics.

5. Conclusions

The analysis demonstrates a clear connection between market sentiment and trading behavior:

1. Traders are significantly more active during Fear sentiment.
2. Trading volume and participation strongly favor Fear periods.
3. Traders achieve higher win rates and more consistent outcomes during Fear.
4. Greed periods result in limited activity, lower performance, and less reliable profitability.
5. Fear-driven volatility appears to be a major catalyst for trader engagement.

For trading strategy development, these findings suggest that:

- Models should account for sentiment conditions.
- Volatility-driven signals may be more effective during Fear.
- Risk controls may be particularly important during Greed, where returns are less predictable.
- Monitoring sentiment cycles could enhance trade timing and risk-adjusted outcomes.

Final Notes

All plots referenced in this report are available in the outputs/ directory.
The full analysis workflow, including data preparation and EDA, is reproducible through the provided notebook.